



近期科学进展

李文楷

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国内外近期科学进展

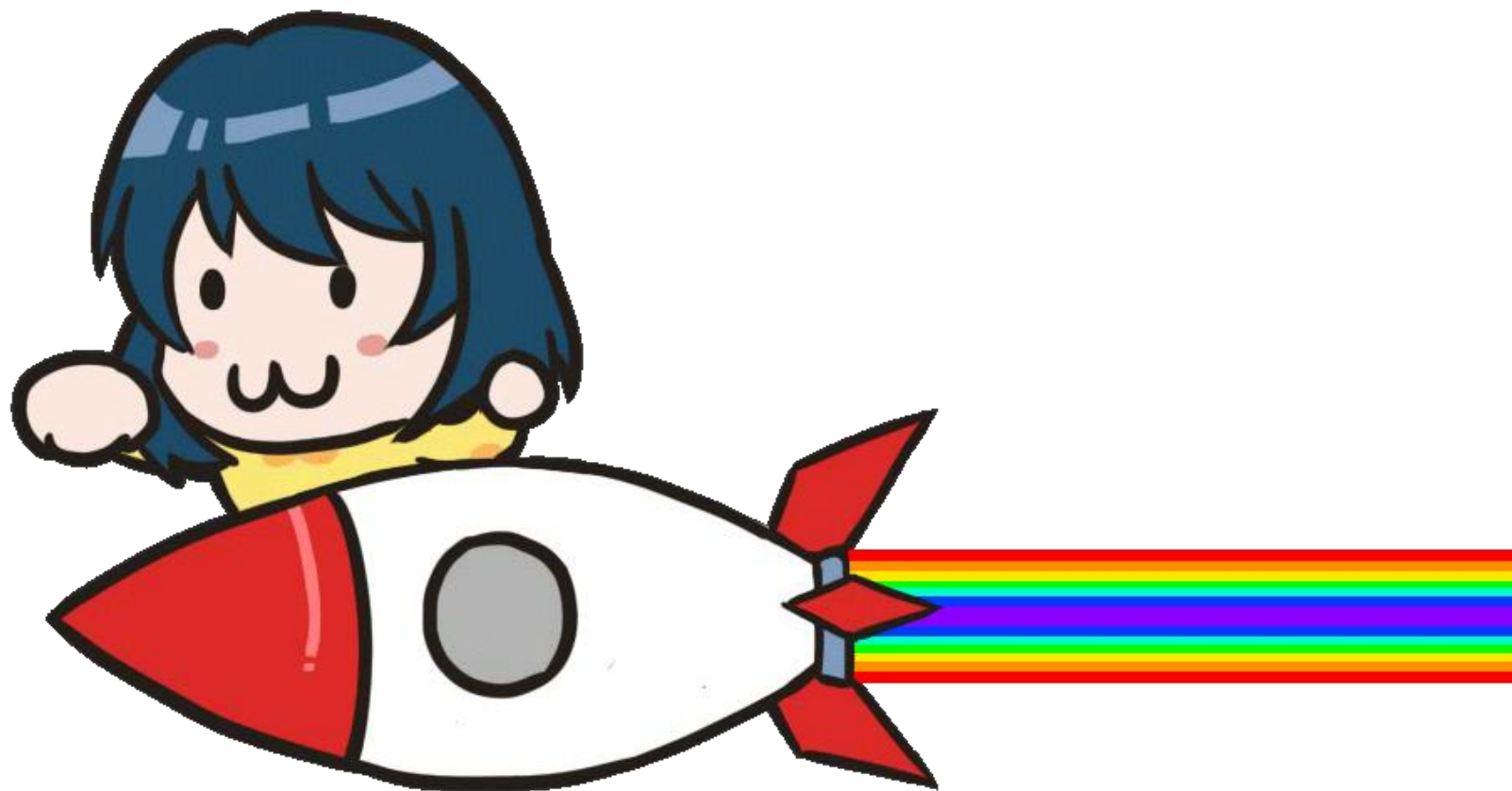


Image credit: 庄晓雨

迄今未知物理起因的超亮暂现源



Very bright optical transient near the Trifid and Lagoon Nebulae

ATel #11448; *Peter Dunsby (University of Cape Town)*
on 20 Mar 2018; 09:14 UT

Distributed as an Instant Email Notice Transients

Credential Certification: Patrick Woudt (pwoudt@ast.uct.ac.za)

Subjects: Transient

Referred to by ATel #: 11449



Twet



Recommend 175

- 1等以上，超亮暂现源
- 80mm口径望远镜发现
- SDSS巡天无已知对应体
- 后随观测发现其自行极大

Peter Dunsby (University of Cape Town) reports the detection of a very bright optical transient in the region between the Lagoon and Trifid Nebulae based on observations obtained from Cape Town on 20 March 2018, between 01:00 and 03:45 UT. The object was visible throughout the full duration of the observations and not seen when this field was observed previously (08 March 2018). The optical transient is at least first magnitude and is located at the following coordinates: RA (2000): 18h 04m 50s Declination (2000.0): -23d 29m 58s The coordinates are accurate to a few arcseconds. There is no obvious counterpart at this position on the Digital Sky Survey plates. Observations were obtained using an 80mm refractor. The attached URL show the image of this field (2.3 x 1.7 degrees, plate scale of 9 arcseconds per pixel) on 20 March 2018. The optical transient is the brightest star in the field. Further observations are strongly encouraged to establish the nature of this very bright optical transient.

Image credit: 庄晓雨



哈勃危机解除，通过地月退行测量 H_0

[Submitted on 30 Mar 2022]

Worry No More, The Hubble Tension is Relieved: A Truly Direct Measurement of the Hubble Constant from Mooniversal Expansion

Gagandeep S. Anand, Zachary R. Clayton, Ryan Dungee

Using sedimentary and eclipse-based measurements of the lunar recession velocity, we derive a new local-Universe measurement of the Hubble constant (H_0) from the recession rate of Earth's Moon. Taking into account the effects of tides, we find a value of $H_0 = 63.01 \pm 1.79 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which is in approximate agreement with the Planck space mission's measurement using the cosmic microwave background (CMB) and base Λ CDM. Our new measurement represents the first ever model-independent, single-step measurement of the Universe's current expansion rate. This is also the first major local Universe measurement of H_0 which is below the measurement from Planck. Importantly, it is robust to the systematic errors that may be present in other H_0 measurements using other cosmological probes such as type Ia supernovae, baryon acoustic oscillations, or lensed quasars. Our work provides key evidence towards the notion that the existing Hubble tension may indeed be a result of systematic uncertainties in the local distance ladder.

Comments: Main result can be seen in Figure 1

Subjects: **Cosmology and Nongalactic Astrophysics** (astro-ph.CO)

Cite as: [arXiv:2203.16551](https://arxiv.org/abs/2203.16551) [astro-ph.CO]

(or [arXiv:2203.16551v1](https://arxiv.org/abs/2203.16551v1) [astro-ph.CO] for this version)

<https://doi.org/10.48550/arXiv.2203.16551> 

Submission history

From: Gagandeep Anand [[view email](#)]

[v1] Wed, 30 Mar 2022 18:00:01 UTC (3,010 KB)

- 首个模型无关的，单步骤的哈勃常数精确测量

$$H_0 = 63.01 \pm 1.79 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

- 严格考虑了地月潮汐作用的影响
- 对现有宇宙模型提出极大挑战
- CMB测量和超新星测量都存在巨大问题?
- 新物理?

Gagandeep S. Anand et al. 2020

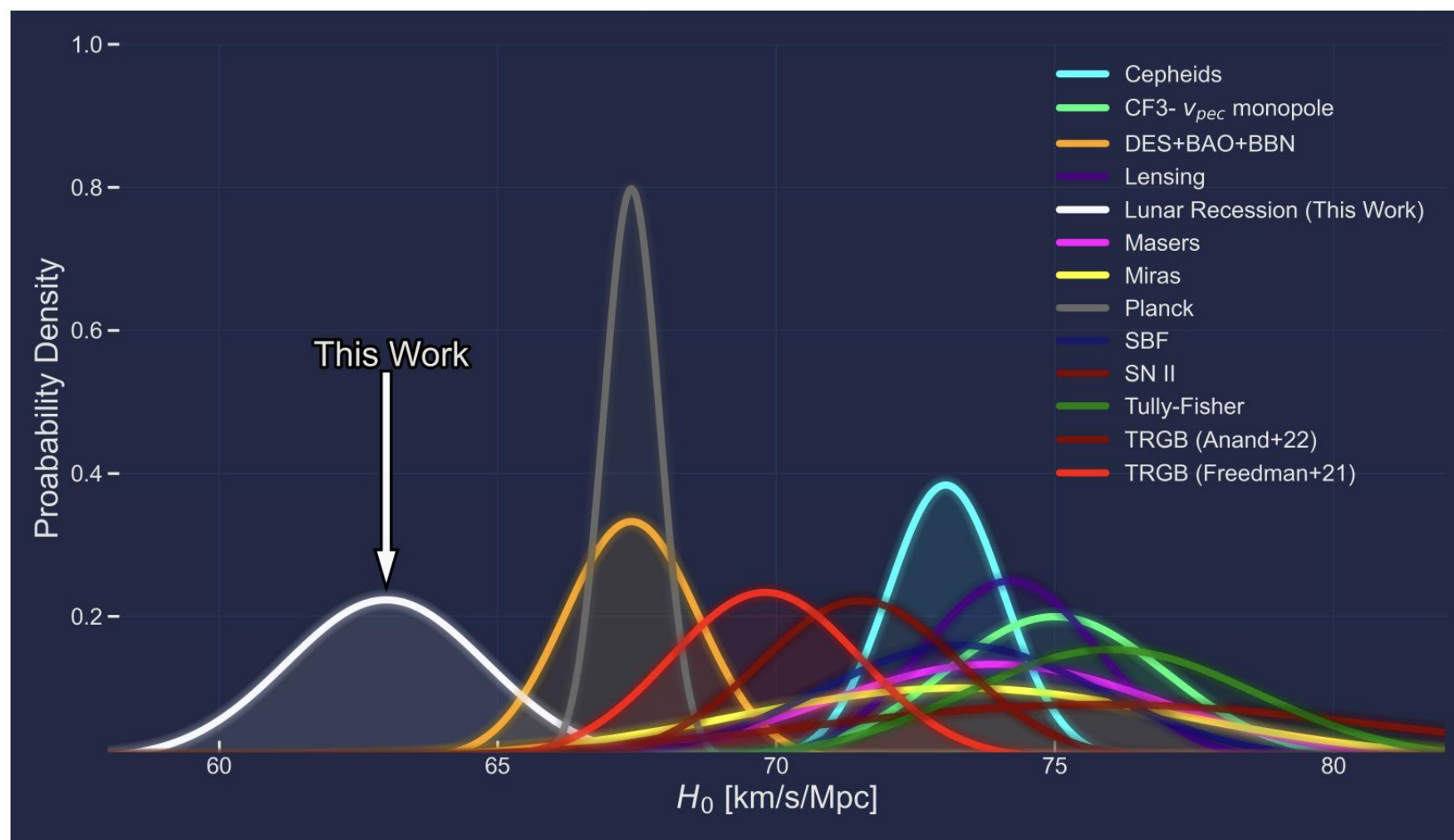


Figure 1. A comparison of numerous recent H_0 results, including the one we derive in this paper, which we stress is the *only* measurement which is both model-independent and performed in a single step. Methods plotted include Cepheid variables (Riess et al. 2021), a restriction on the peculiar velocity monopole from Cosmicflows-3 (Tully et al. 2016), the Dark Energy Survey + Baryon Acoustic Oscillations + Big Bang Nucleosynthesis (Abbott et al. 2018), time delay cosmography of lensed quasars from TDCOSMO (Millon et al. 2020), megamasers from the Megamaser Cosmology Project (Pesce et al. 2020), Mira variables (Huang et al. 2020), Planck measurements of the CMB (Planck Collaboration et al. 2020), surface brightness fluctuations (Blakeslee et al. 2021), type II supernovae (de Jaeger et al. 2020), the Tully-Fisher relation (Kourkchi et al. 2020), and the Tip of the Red Giant Branch (Freedman 2021; Anand et al. 2021).

Gagandeep S. Anand et al. 2020

WFST暑校期间科学进展

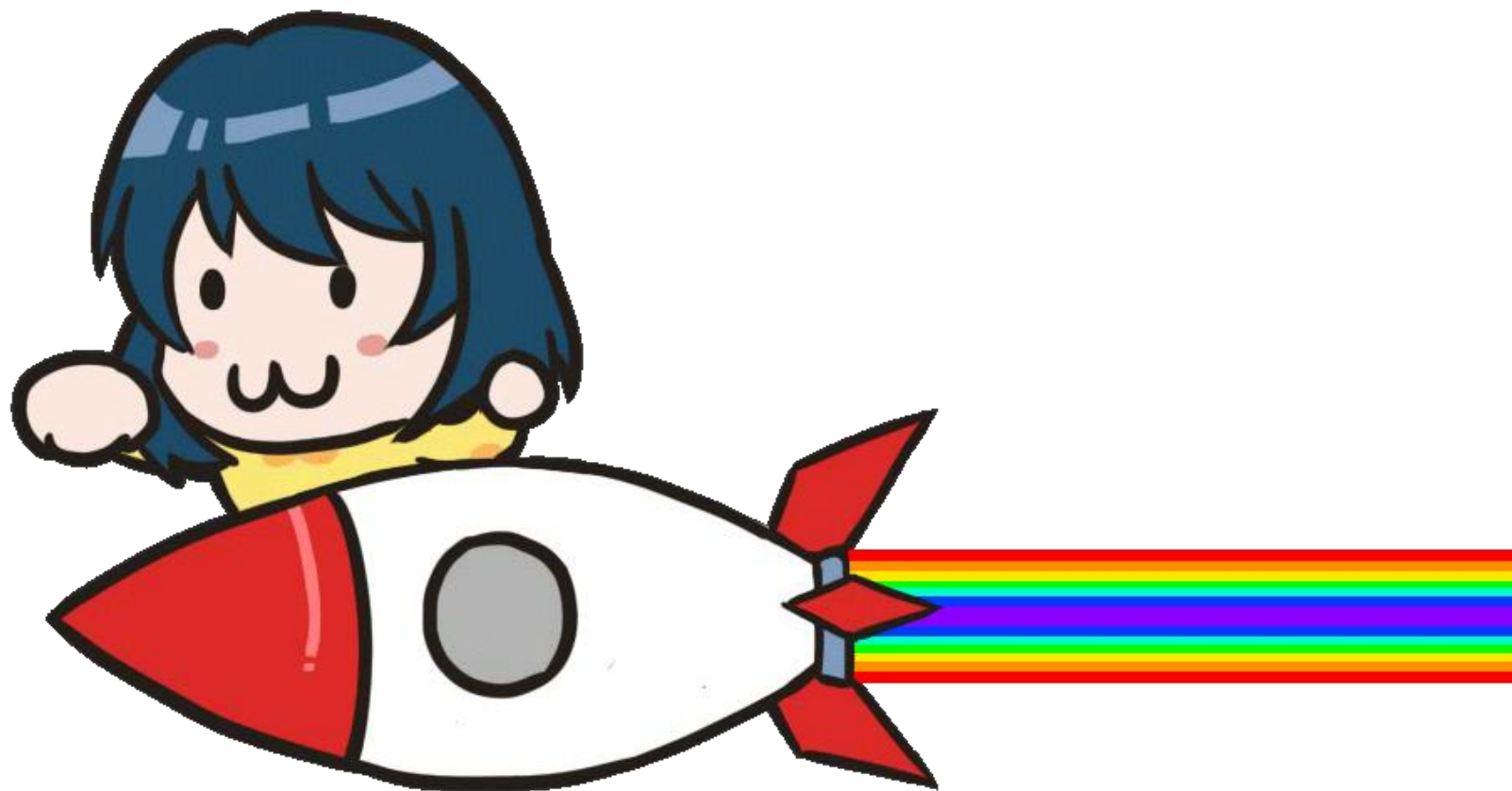
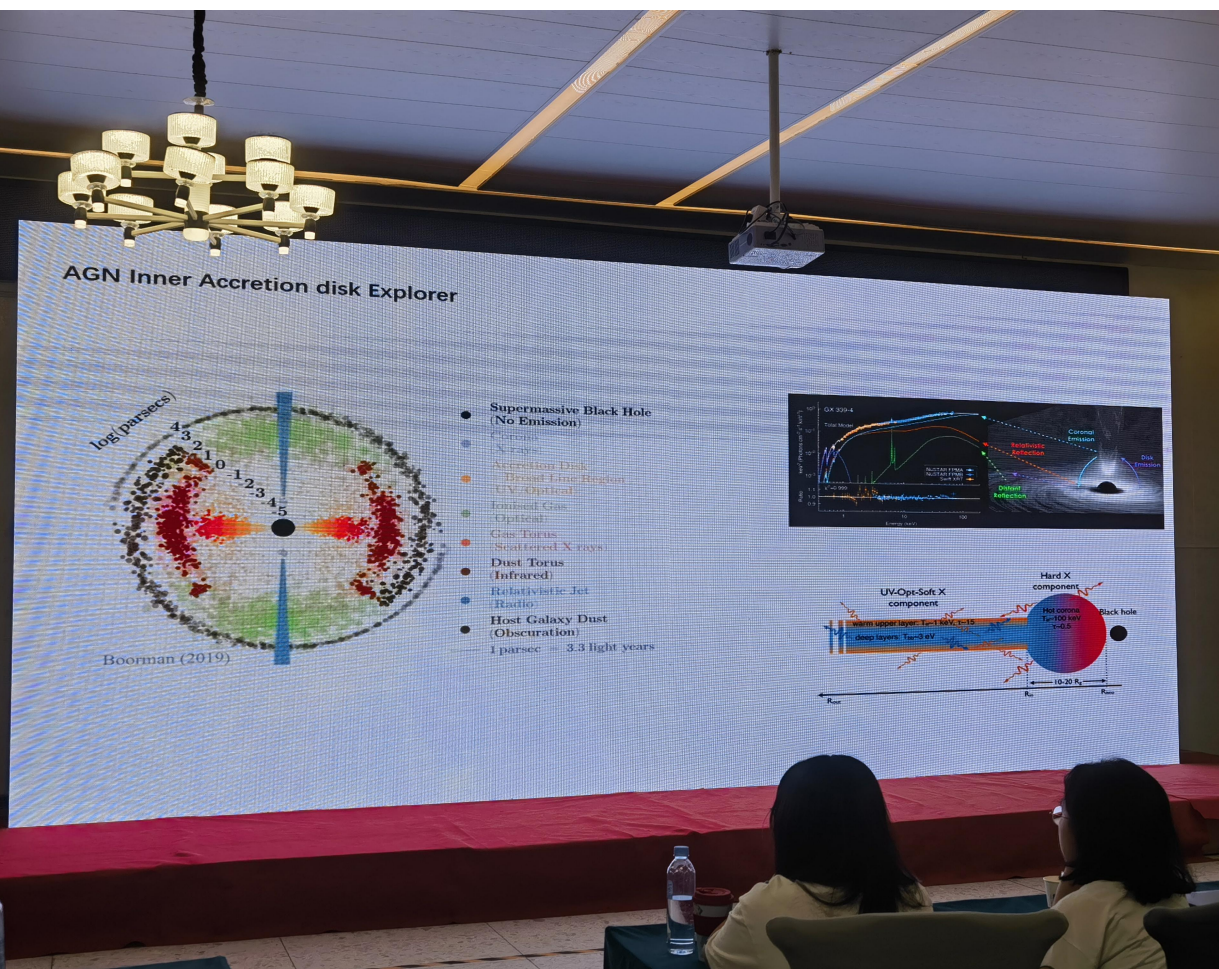


Image credit: 庄晓雨

AGN Inner Accretion Disk Explorer (PI: 陈世江)



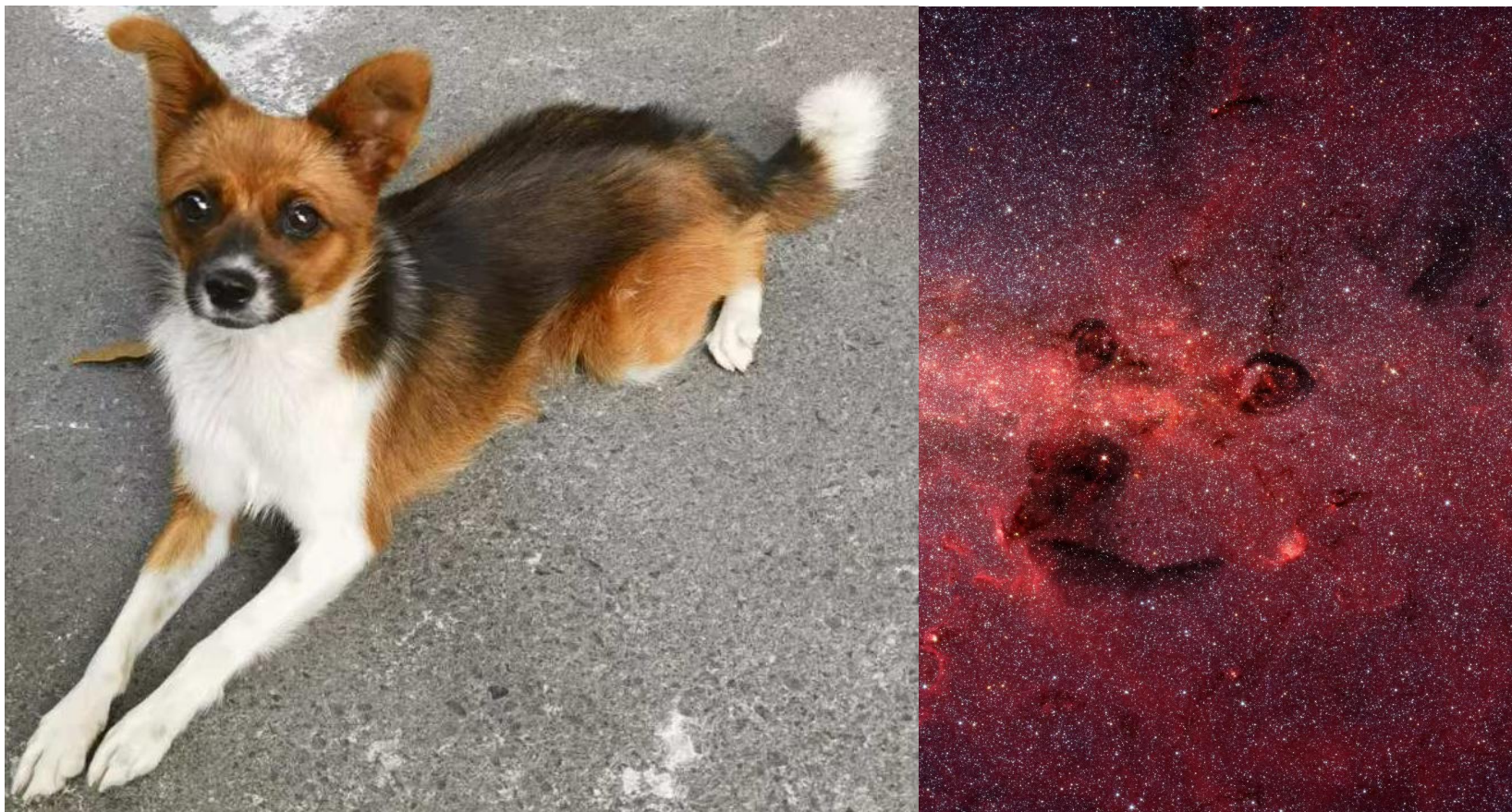
攻克宏观量子克隆技术难关



Image credit: 庄晓雨

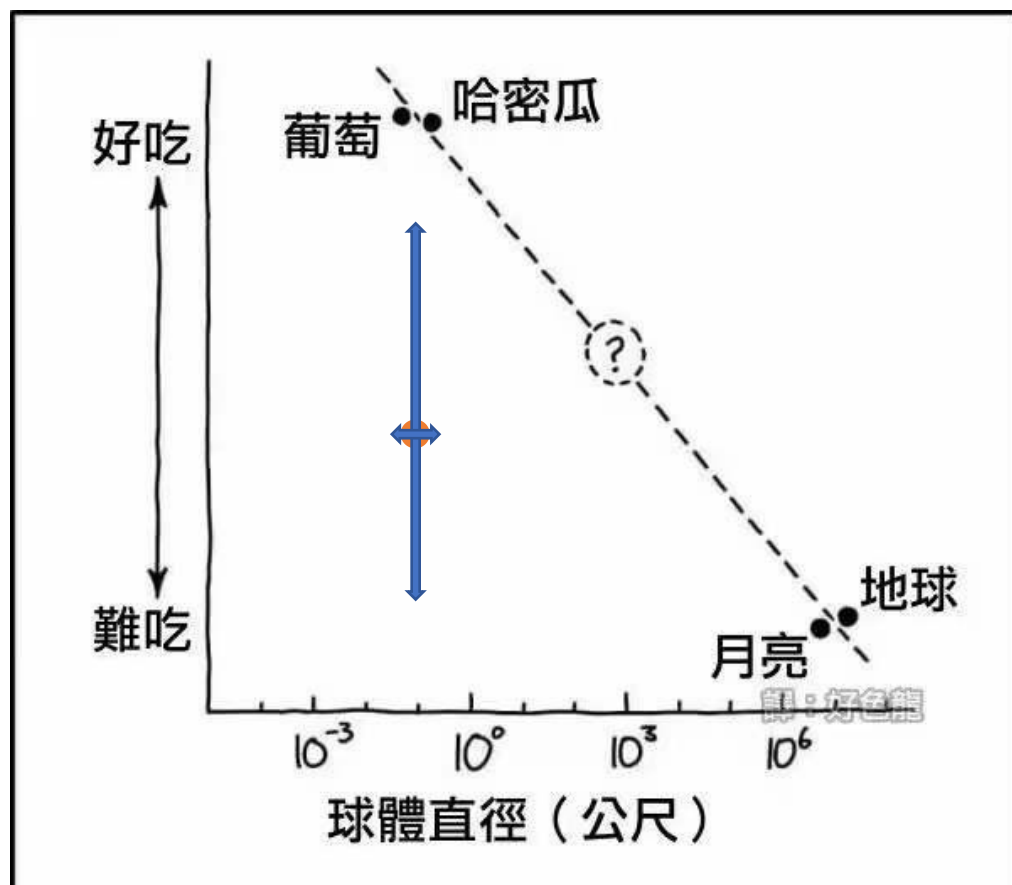


拓展量子克隆技术，成功将 克隆至宇宙尺度



- 此技术尚有缺陷
- 无法完美复制
- 持续时间较短
- 未来应用前景广泛
- ~~整个宇宙为你闪耀~~

离群点，严重挑战既有定律。暗示着新物理？



我的研究表示這個世界上可能存在著一個直徑800公尺、味道還可以的球體

- “合成参数”，反应总体的统计性质
- “无偏”样本（身边统计学，来自同学和师兄）
- 可靠的误差估计（统计误差+本征弥散）
- 存在自相似解？后续进一步研究



踝区神秘鼓包，暗示太阳周日运动

- 明显的成团性（特征尺度**2cm**）
- 耀发后，呈现先上升再下降的光变特征
- 目前仍处于**peak**，需要后续观测
- 多发于观星后，强烈暗示天体物理起源
- 多发于夜晚，强烈暗示太阳起源



教室内的密度波：时间和空间上成协？

- 目前有四个此类事件（按能量释放效率分为两类），样本容量仍不够大
- 有待进一步探究
- 有一定的选择效应没有扣除

Image credit: 庄晓雨



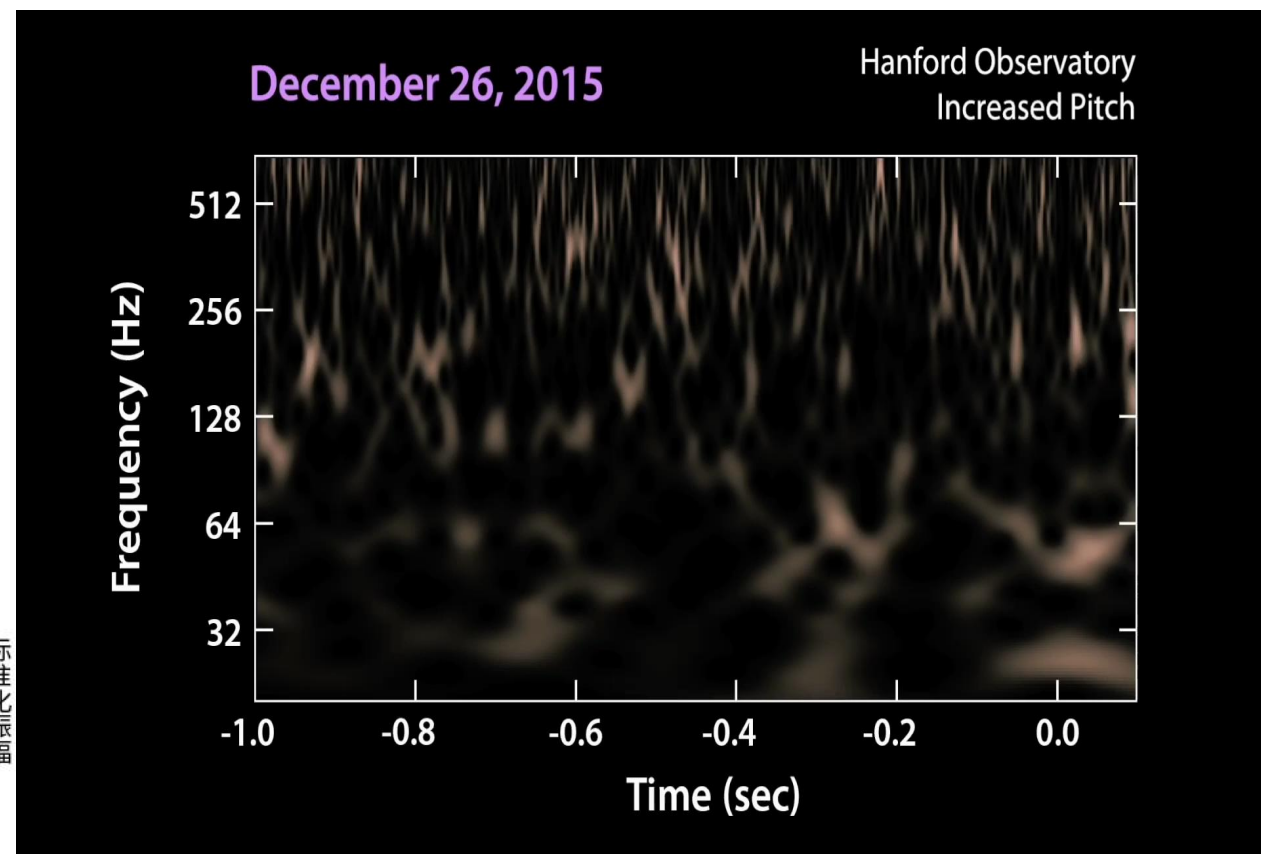
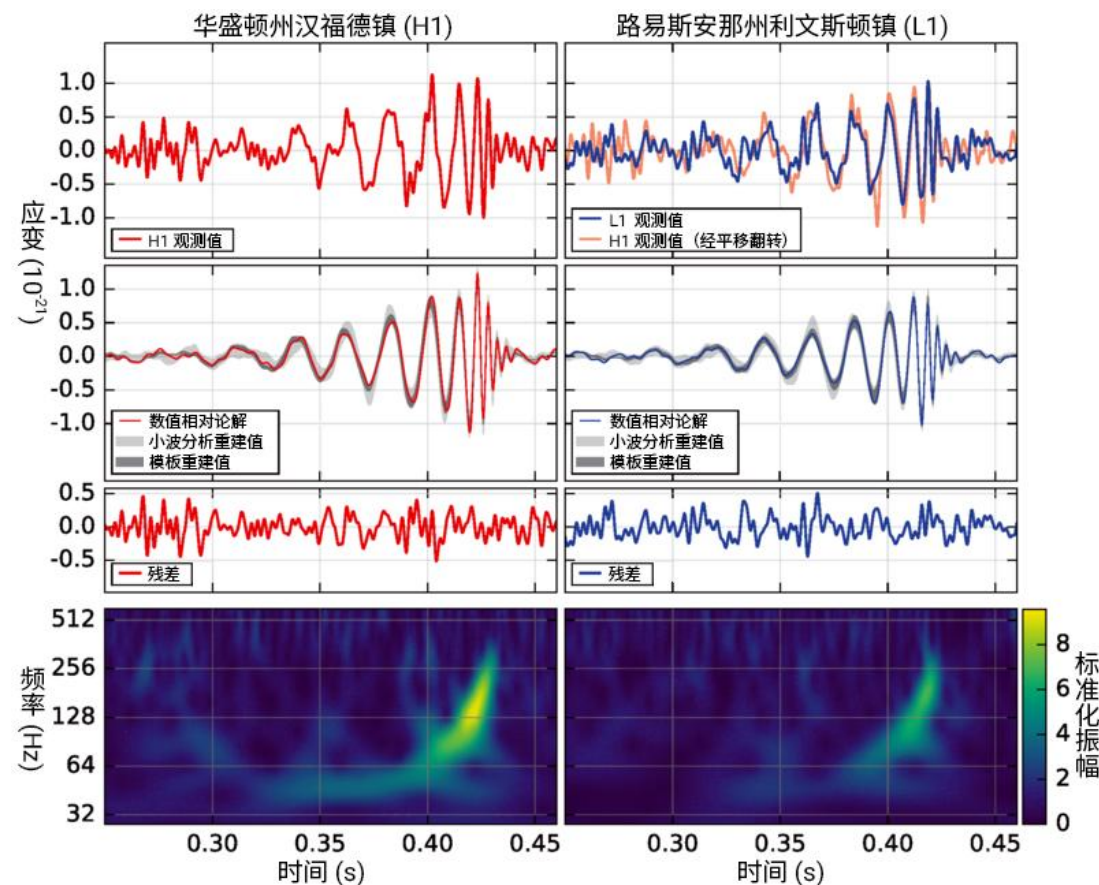
/zhōujiū/

拟声词 文言

chirp mass

模拟鸟叫的声音。
「乳雀 啾啾」

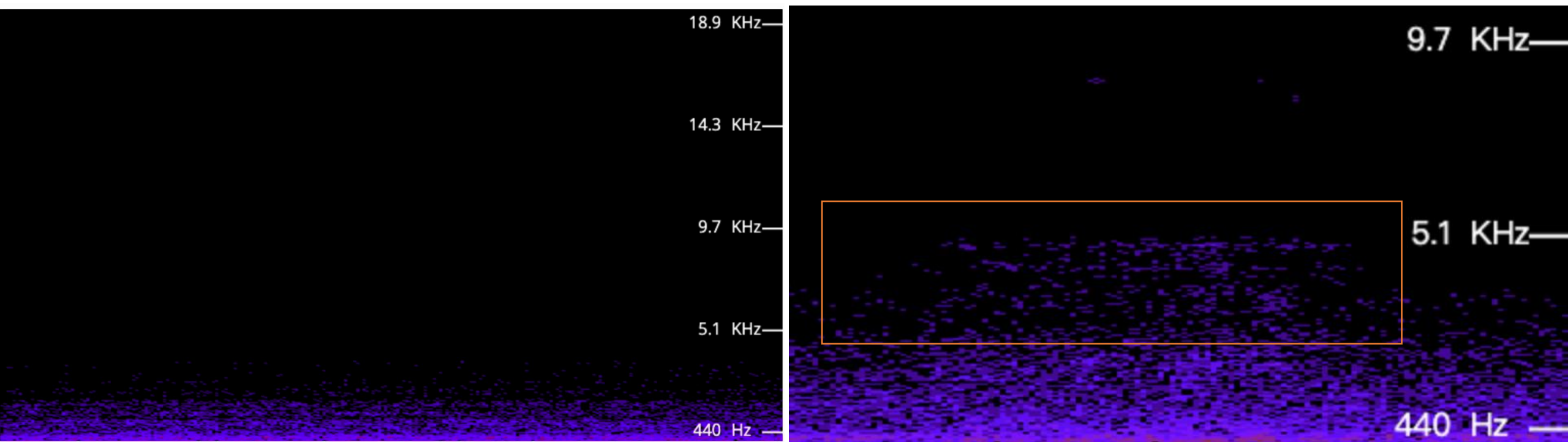
2017: 引力波直接探测





2024WFST暑校：迄今为止最小质量中子星的发现

- 存在短时标的周期性（约3–4s的周期）→宁静态，能谱较软
- 长时标的周期性耀发？ 3h内观测到了3次耀发，呈现准周期性→爆发态，能谱较硬，持续5s–1min
- 覆盖区间不是很完善，没有凝视观测
- 观测资源比较有限，观测助手同时承担天协ddl，表演节目准备等多项任务



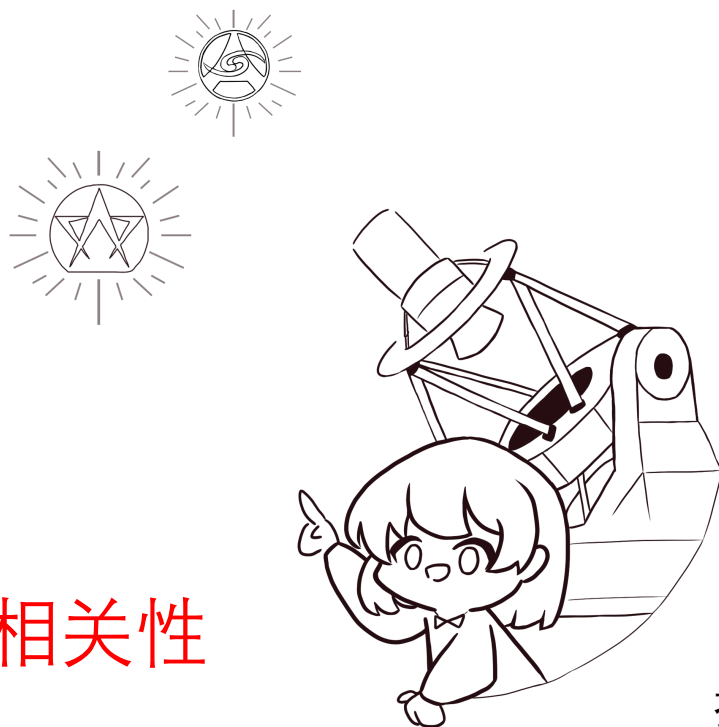
大样本结论：改名姓王，天文就业新思路

教授、研究员：[常进](#) [蔡一夫](#) [程福臻](#) [褚耀泉](#) [戴子高](#) [范璐璐](#) [方文娟](#) [冯畅](#) [Filipe Batoni Abdalla](#) [何志成](#) [姜继安](#) [孔旭](#) [李剑](#) [刘桂琳](#) [刘腾](#) [容昱](#) [王恩赐](#) [王慧元](#) [王俊贤](#) [王挺贵](#) [向守平](#) [薛永泉](#) [严飞](#)
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讲师：[陈次星](#) [王春成](#) [杨建明](#)

- 全国约1亿王姓人口
- 发生率 $< \sim 10\%$
- 中科大天文系王姓老师： $7/40=17.5\%$
- WFST暑校王姓老师： $3/6=50\%$
- 显著高于全国平均值！ $\rightarrow$ 暗示极强的相关性



指导老师：姜继安

多信使占星术： 天文就业新思路

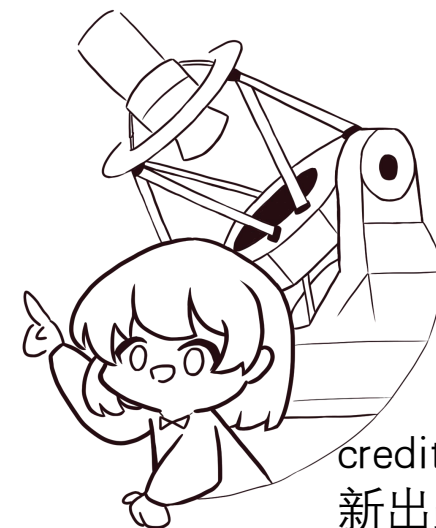
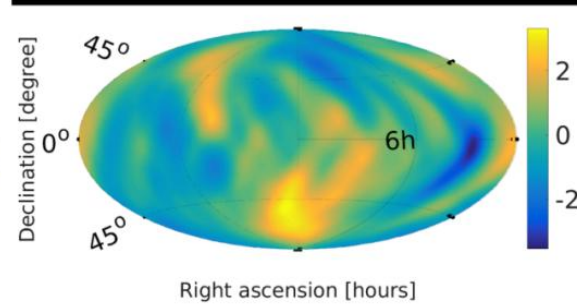
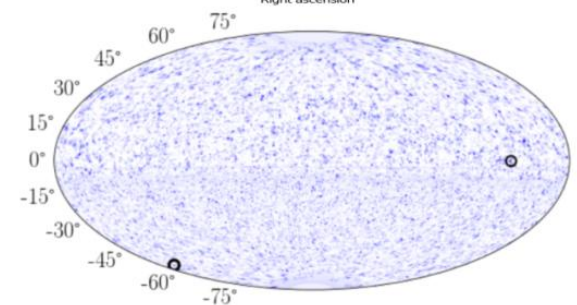
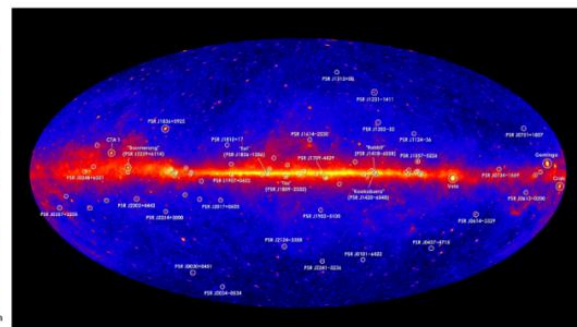
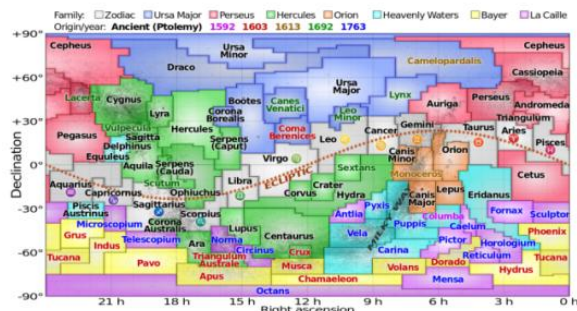
[Submitted on 28 Mar 2024]

Multi-Messenger Astrology

Gwen Walker, Nick Ekanger, R. Andrew Gustafson, Sean Heston

It has long been accepted that the cosmos determine our personalities, relationships, and even our fate. Unlike our condensed matter colleagues – who regularly use quantum mechanics to determine the healing properties of crystals – astrology techniques have been unchanged since the 19th century. In this paper, we discuss how astrophysical messengers beyond starlight can be used to predict the future and excuse an (1) fraction of our negative personality traits.

基于来自Astrological Sign Research Institute、Space Vision Research Center、冰立方中微子观测实验室、激光干涉引力波天文台、Center for Astrological Personality Traits Research、Center for Extraterrestrial Biology Research等多个团组的数据，**和好事天文学家的共同努力**，多信使占星预测的研究已初具雏形。

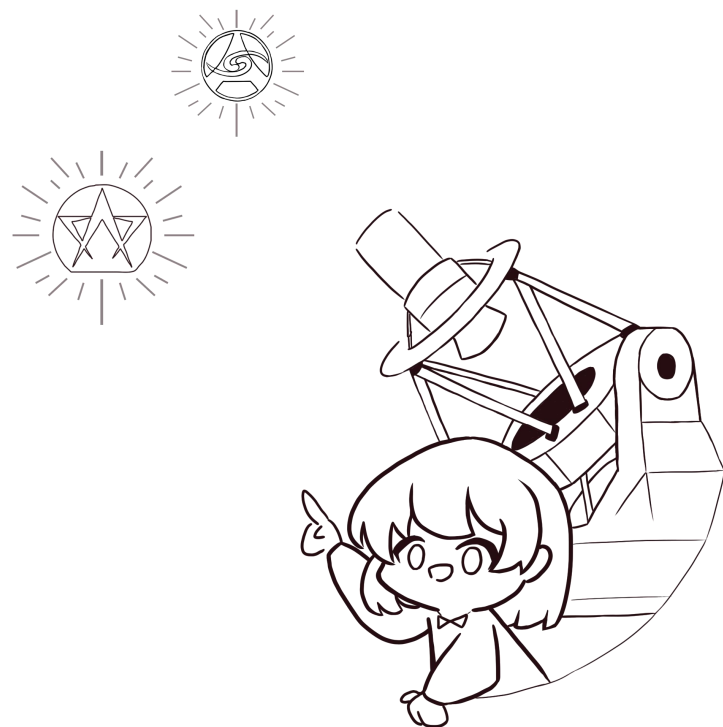


credit: 《震惊！天文科研狗的新出路，速进！》，蜗壳天文

多信使占星术：伽马波段



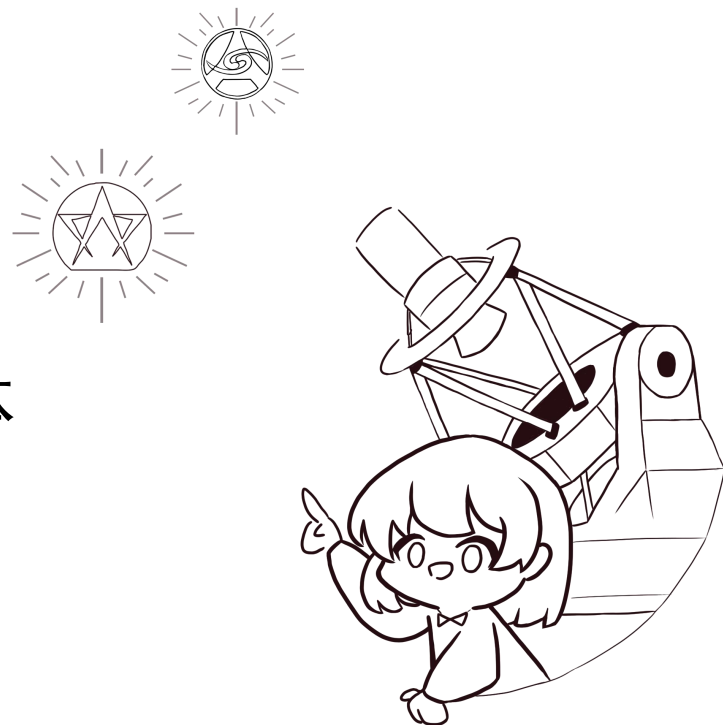
- 被伽马射线过量照射→易怒、情绪问题
- 某星座有持续的伽马射线源→性格不稳定的概率会增加



credit: 《震惊！天文科研狗的新出路，速进！》，蜗壳天文

多信使占星术：宇宙线

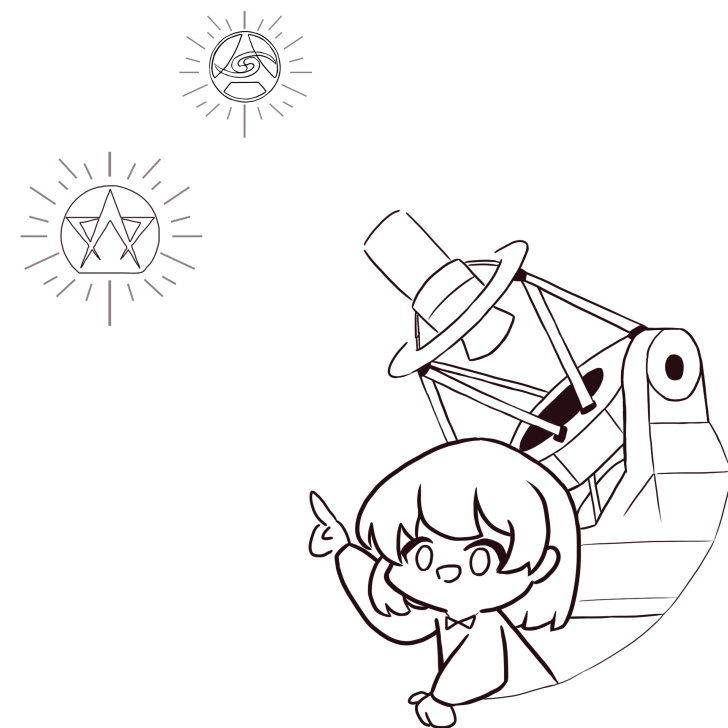
- Gwen Walker等人引入了一个名为“棒球速度 (Baseball Speed, BS)”的新单位
- 一个投掷出的棒球的平均能量 ($\sim 50\text{J}$)
- 显然，一名出色的棒球运动员可以作为一个天体物理源的加速器，将物体加速到1 BS
- 尽管名称中有“速度”一词，但BS是一个能量单位



credit: 《震惊！天文科研狗的新出路，速进！》，蜗壳天文

多信使占星术：中微子

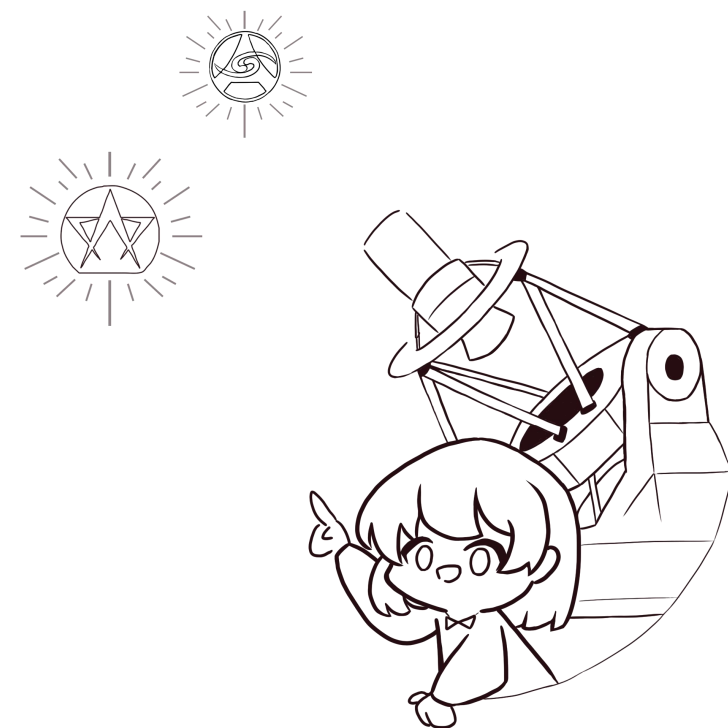
- 也被称为幽灵粒子
- 像幽灵一样很少与其他物质相互作用
- 在中微子下出生的人有更高的灵性感知，并且更有可能鬼鬼祟祟地消失
- 中微子的主要来源之一是太阳，当太阳升起时，中微子的涌入可能导致论文作者莫名从约会对象面前消失



credit: 《震惊！天文科研狗的新出路，速进！》，蜗壳天文

多信使占星术：引力波

- 引力波是最新的占星术信使
- 在占星学框架中，引力波决定了一个人的吸引力和ta对他人的吸引力
- （这“显而易见”；聪明人无需过多解释）



credit: 《震惊！天文科研狗的新出路，速进！》，蜗壳天文

举个例子：白羊座

- 有伽玛射线信号
- 有热情、积极进取、追求公平正义
- 持续存在的中微子源
- 神出鬼没
- 人际关系可能会更加脆弱
- 潜在的引力波空白区域之内
- 仍需进一步量化

星座分析

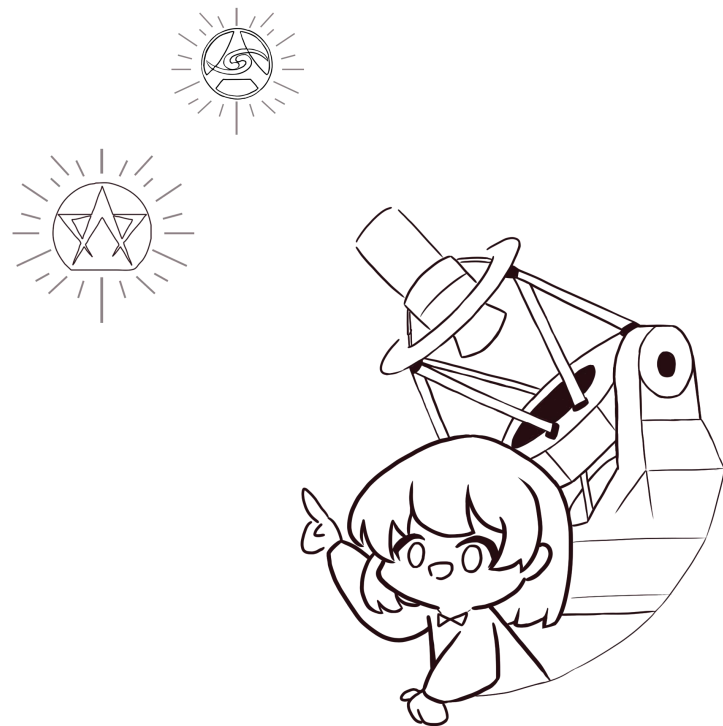
你是个早起打工人



credit: 《震惊！天文科研狗的新出路，速进！》，蜗壳天文

总结与展望

- 权威机构数据
- 样本(人类)充足
- 前景可期
- 已经能较好解释人类个体的一些“缺陷”
 - 如：不回复邮件，发放工资意愿匮乏，拒绝别人精心提出的观测申请，因理解他人文字的能力欠佳而拒稿等等
- 该方向还有待改进之处
 - 同行评议的标准难以统一
 - 被评估样本可能会出现负面情绪
 - 难以获得基金支持等



credit: 《震惊！天文科研狗的新出路，速进！》，蜗壳天文

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江苏

由中国科学技术大学天文学系主办、学生主导的天文科普平台 >

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作者精选

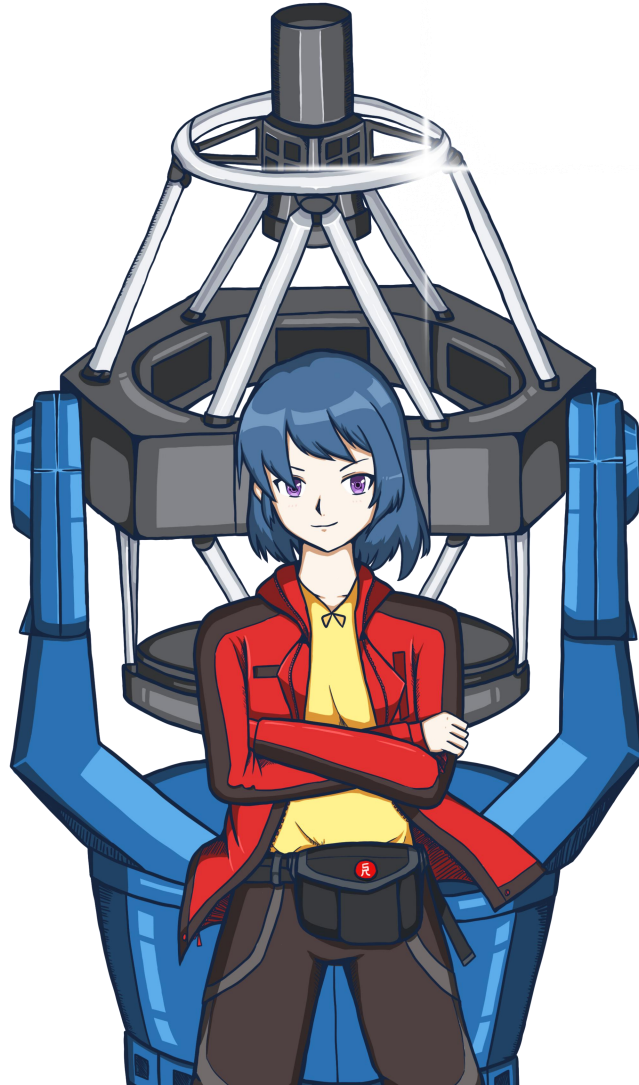
墨子巡天望远镜安装调试初步历程
阅读 1743 赞 37



昨天

两颗中子星碰撞会发生什么？探秘致密双星共舞

阅读 342 赞 19 1个朋友在看



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- 1、语言流畅，通俗易懂，科学严谨，无明显错误，适当结合天文时事热点；
- 2、要求稿件为原创作品，对文字、图片、视频的引用应当注明出处（网站链接），**无版权纠纷**；
- 3、尽可能通过图片、视频等方式阐释科学知识，避免公式和晦涩的学术用语；
- 4、投稿后由编辑小组成员负责审核、校对和修改，直至定稿。**作者应配合编辑完成稿件的修改工作。**

300元/千字



稿酬

投稿详情联系：吴薇羽

不低于业界标准，以定稿篇幅和类型计算基础稿酬，此外根据阅读量增加创作激励奖金。

参考酬劳：300元/千字(专业科普)。

文章在公众号发表后结算。

Image credit: 庄晓雨